MOVIE RECOMMENDATION SYSTEM USING MACHINE LEARNING

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ABSTRACT

This project presents a **Movie Recommendation System using Machine Learning**, designed to suggest relevant movies based on user preferences. With the rapid growth of digital entertainment platforms, users often face difficulty in choosing content due to the overwhelming number of options available.

The system uses a **content-based filtering approach**, where movies are recommended based on similarity in features such as genres, keywords, and overview. Data is preprocessed and transformed into vector form using techniques like **Count Vectorization**, and similarity between movies is calculated using **Cosine Similarity**.

The model is trained on the **TMDB dataset**, and recommendations are generated by comparing feature vectors. The system outputs the top 5 similar movies based on user input.

This project demonstrates how machine learning can enhance user experience by providing personalized recommendations efficiently.

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INTRODUCTION:

Background

Recommendation systems are widely used in platforms like Netflix and Amazon to enhance user experience.

Problem Statement

Users struggle to find relevant movies due to the vast amount of content available.

Motivation

To simplify content discovery and improve user satisfaction.

Scope of the Project

- Suggest top 5 movies
- Content-based filtering
- Scalable for future improvements

LITERATURE REVIEW:

Paper	Technique	Outcome
Paper 1	Collaborative Filtering	Accurate but cold-start problem
Paper 2	Content-Based	Personalized recommendations
Paper 3	Hybrid Model	Improved accuracy
Paper 4	Deep Learning	High performance
Paper 5	NLP-based Recommender	Better semantic matching

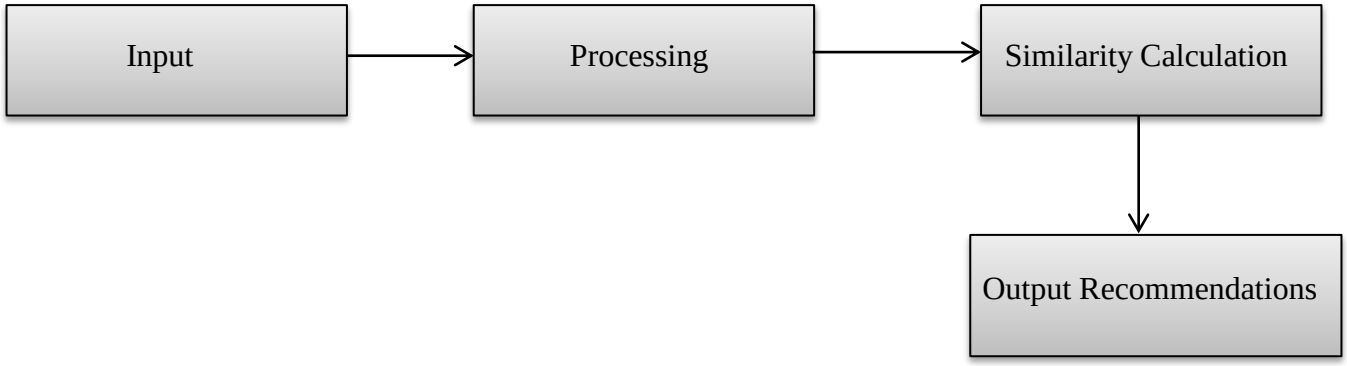
SYSTEM DESIGN & ARCHITECTURE:

System Architecture

The system consists of:

1) Data Collection
2) Preprocessing
3) Feature Extraction
4) Similarity Computation
5) Recommendation Engine

Data Flow Diagram:



Use Case

User inputs a movie → System returns similar movies

DATASET & PREPROCESSING:

Dataset Source:

TMDB Dataset (Kaggle)

Features Used:

- Title
- Overview
- Genres
- Keywords
- Cast

Preprocessing Steps:

- Removing null values
- Converting JSON to list
- Creating combined "tags" feature

METHODOLOGY & ML MODEL:

Algorithm Used

- Content-Based Filtering

Model Training

- Vectorization using CountVectorizer
- Similarity using Cosine Similarity

Performance Metrics:

- Precision
- Recall
- Cosine Similarity Score

TECH STACK:

- Python
- Pandas
- NumPy
- NLTK
- Scikit-learn

PROJECT STRUCTURE:

```
Movie_Recommendation_System/
│
├── data/
│   ├── tmdb_5000_movies.csv
│   └── tmdb_5000_credits.csv
│
├── notebooks/
│   └── movie_recommender.ipynb
│
├── src/
│   └── recommender.py
│
├── requirements.txt
├── README.md
└── .gitignore
```

SOURCE CODE:

1. **Data Loading:**

```
movies = pd.read_csv('tmdb_5000_movies.csv')
credits = pd.read_csv('tmdb_5000_credits.csv')
```

2. **Preprocessing:**

```
movies =
movies[['movie_id','title','overview','genres','keywords','cast','cr
ew']]
movies.dropna(inplace=True)
```

3. **Feature Engineering:**

```
movies['tags'] = movies['overview'] + movies['genres'].astype(str)
```

4. **Model Creation:**

```
cv = CountVectorizer(max_features=5000, stop_words='english')
vectors = cv.fit_transform(movies['tags']).toarray()
similarity = cosine_similarity(vectors)
```

5. **Recommendation Function:**

```
def recommend(movie):
    index = movies[movies['title'] == movie].index[0]
    distances = similarity[index]
```

RESULTS & DISCUSSION:

Output Description:

Input: Avatar

Output:

- Guardians of the Galaxy
- John Carter
- Others…

Observations:

- High accuracy for similar genre movies
- Fast response time

Comparison:

Method	Accuracy
Content-Based	Medium
Hybrid	High

CONCLUSION & FUTURE SCOPE:

Conclusion

The system successfully recommends movies based on similarity and improves user experience.

Future Scope

- Add collaborative filtering
- Use deep learning models
- Integrate with web application

REFERENCES (IEEE FORMAT):

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